

FLOWMIND AI CAPSTONE PROJECT

Week-Wise Development Plan & 4-Teammate Work Allocation

Software Development Life Cycle (SDLC) Execution & Task Responsibility Breakdown

1. Executive Summary & Team Domain Allocation

This plan defines the 8-week execution strategy for building **FlowMind AI**, focusing on delivering a layman-accessible **Chatbot MVP by Week 3**, followed by complete enterprise business modules, 15 AI agents, workflow automation, data studio, RAG knowledge base, and proactive operational monitoring.

Teammate 1: AI & Multilingual Lead

Focus: LLM Orchestration, 15 Specialized AI Agents Mesh, Indic Script Speech/NLP Engine (EN, HI, TE, TA, KN, ML), Prompt Protection.

Teammate 2: Frontend & Layman UI/UX Lead

Focus: Next.js 14 App Router, Glassmorphic UI System, Layman Chatbot MVP, Agent Thought Streamer, Action Cards, Audio Waveforms.

Teammate 3: Workflow & Business Engine Lead

Focus: NL-to-DAG Workflow Builder, Execution DAG Runner, Business Suite (CRM, HR, Finance, Inventory, Support), Proactive Monitor.

Teammate 4: Data, RAG & Security Infrastructure Lead

Focus: Interactive Data Studio (CSV/PDF Parser & Chart AI), Vector RAG Knowledge Base, DB Schemas, Auth, RBAC & Multi-Tenancy.

2. 8-Week Master Project Roadmap

Week	Phase Goal	Key Deliverable	Milestone Target
Week 1	SDLC Setup & Design System	Repo, Layout, Design Tokens, Database Schemas	Architecture Sign-off
Week 2	Multilingual & Chatbot Foundation	Indic Language Router, Chat UI, Voice Spectrum	Multi-script Text & Speech
Week 3	15-Agent Mesh & Approval Cards	Thought Stream Visualizer, Action Confirmation Cards	Milestone 1: Chatbot MVP

Week	Phase Goal	Key Deliverable	Milestone Target
Week 4	NL Workflow Automation Engine	Natural Language DAG Builder, Execution Runner	Zero-Code Workflows
Week 5	Business Operations Suite	CRM, HR, Finance, Inventory, Support Modules	Integrated Business Suite
Week 6	Data Studio & RAG Knowledge Base	CSV/PDF Auto-cleaner, Chart AI, Document Citation Q&A	Milestone 2: Data & Knowledge Hub
Week 7	Proactive AI & Security Hardening	Proactive Anomaly Monitor, RBAC, Multi-tenancy	Security & Health Audit
Week 8	System Integration & Presentation	End-to-End Verification, Performance Tuning, Demo	Final Production Sign-off

3. Detailed Week-by-Week Development Breakdown

Week 1: SDLC Architecture, Design System & Foundation Setup

Goal: Technical architecture approved & design tokens created

- **Teammate 1:** Setup LLM SDK connectors, define unified canonical business intent JSON schema, research Indic script tokenization.
- **Teammate 2:** Initialize Next.js 14 with TypeScript & Tailwind CSS, build layout frame (Sidebar, Topbar, Language Switcher), set up Glassmorphism token CSS.
- **Teammate 3:** Define Workflow DAG JSON schema, create mock data layer for enterprise entities (Leads, Invoices, Employees), design state machines.
- **Teammate 4:** Design PostgreSQL/SQLite ERD, setup authentication scaffolding (JWT, RBAC roles), configure repository environment & CI rules.

Week 2: Multilingual Engine & Layman Chatbot Foundation

Goal: Interactive multi-language chat interface live

- **Teammate 1:** Build Multilingual Router for English, Hindi, Telugu, Tamil, Kannada, and Malayalam; implement language auto-detection & TTS voice mapping.
- **Teammate 2:** Build Layman Chatbot page (`/chat`), implement interactive sample prompt chips in native scripts, create Audio Waveform spectrum visualizer.
- **Teammate 3:** Build intent extraction logic for core actions (Create Lead, Show Sales, Draft Invoice), set up real-time execution log structures.
- **Teammate 4:** Build Chat History & Session APIs, implement Multi-tenancy Organization isolation middleware, setup Redis conversation cache.

Week 3: 15-Agent Mesh Collaboration & Chatbot MVP Launch

MILESTONE 1: Layman Chatbot MVP Live

- **Teammate 1:** Implement Master Agent Orchestrator across 15 specialized agents, write prompts for Sales, Finance, HR, and Analytics agents, add prompt injection filters.
- **Teammate 2:** Build Agent Thought Stream visualizer (shows live step progress pills), build Action Confirmation Cards with [Approve & Execute] / [Edit] / [Cancel] buttons.
- **Teammate 3:** Connect agent routing output directly to Action Cards, implement quick approval execution flow, create background notification toasts.
- **Teammate 4:** Build audit logging service for agent decisions and user approvals, optimize conversation memory window context.

Week 4: Natural Language Workflow Automation Engine

Goal: Zero-code NL DAG builder and runner functional

- **Teammate 1:** Prompt/train Workflow Planner Agent to convert plain text into runnable DAG JSON graphs with conditional logic.
- **Teammate 2:** Build Visual Workflow Page (`/workflows`), create interactive DAG node graph view, build Live Execution Runner console.

- **Teammate 3:** Develop Workflow DAG Runner Engine (executes node sequence, handles retries, webhook triggers, scheduled cron).
- **Teammate 4:** Implement persistent Workflow storage in DB, build background job runner / async task queue simulation, setup webhook listeners.

Week 5: Integrated Business Operations Suite

Goal: Operational suite (CRM, HR, Finance, Inventory, Support) functional

- **Teammate 1:** Equip specialized domain agents with business tools, implement automated summarization for weekly sales, payroll, and support.
- **Teammate 2:** Build UI views for CRM, HR, Finance, Inventory, and Support modules (`/modules/\*`) with interactive data tables and quick action drawers.
- **Teammate 3:** Implement core business logic (Lead stage progression, Payroll calculation, Invoice PDF generation, Stock alerts, Support tickets).
- **Teammate 4:** Optimize database query performance for business entities, implement CSV/JSON data export utilities, secure role-based access for HR & Finance.

Week 6: AI Data Studio & Enterprise RAG Knowledge Base

MILESTONE 2: Data Studio & RAG Knowledge Hub Live

- **Teammate 1:** Implement Document RAG Agent (text chunking, vector embeddings, similarity search) and AI Data Analyst Agent (statistical summary & chart recommendations).
- **Teammate 2:** Build Interactive Data Studio (`/data-analyst`) with drag-and-drop uploader, Auto-Generated Chart Studio, and RAG Knowledge Hub (`/knowledge`) with citation links.
- **Teammate 3:** Build client-side CSV/PDF cleaning pipeline (missing value detection, type casting), implement AI Executive Report Generator (`/reports`).
- **Teammate 4:** Implement vector store simulation (pgvector compatible), build document storage API with metadata extraction, ensure multi-tenant document isolation.

Week 7: Proactive AI Monitor, Security (RBAC) & System Integration

Goal: Proactive monitor live & security hardened

- **Teammate 1:** Implement Proactive Insight AI Engine (scans for overdue invoices, declining sales, stock shortages), format alerts into natural language recommendations.
- **Teammate 2:** Build Proactive Insights Page (`/proactive`), 1-Click Action Alerts inside Chatbot MVP, and Enterprise Settings (`/settings`) with Role Switcher & API Key Manager.
- **Teammate 3:** Connect Proactive Monitor triggers to automated workflow suggestions, implement Third-Party Integrations simulation (Slack, Google, Stripe, WhatsApp).
- **Teammate 4:** Enforce Role-Based Access Control (RBAC) across all API routes, implement secret key encryption for API credentials, build audit log viewer.

Week 8: System Integration, Testing & Presentation

FINAL SIGN-OFF: Capstone Demo Ready

- **Teammate 1:** Conduct comprehensive multilingual conversational testing across Telugu, Hindi, Tamil, Kannada, Malayalam, and English.
- **Teammate 2:** Perform UX accessibility audit, ensure responsive mobile layout, add smooth micro-animations for page transitions.
- **Teammate 3:** Execute end-to-end scenario testing (NL Query -> Agent Mesh -> Workflow DAG -> Business Module -> Proactive Alert).

- **Teammate 4:** Run build verification (`npm run build`), generate documentation artifacts, package application for presentation.

4. Work Responsibility Summary Table

Deliverable Module	Primary Lead	Secondary Support	Key Output Artifact
Layman Chatbot MVP	Teammate 2 (UI/UX)	Teammate 1 (AI)	`ChatInterface.tsx`, `ThoughtStream.tsx`, `ActionCard.tsx`
Multilingual Engine	Teammate 1 (AI)	Teammate 2 (UI/UX)	`languageDetector.ts`, `VoiceSpectrum.tsx`, i18n Dictionary
15-Agent Orchestrator	Teammate 1 (AI)	Teammate 3 (Workflow)	`agentOrchestrator.ts`, Agent Mesh Visualizer
NL Workflow Engine	Teammate 3 (Workflow)	Teammate 1 (AI)	`workflowEngine.ts`, `WorkflowBuilder.tsx`, DAG Runner
Business Operations Suite	Teammate 3 (Workflow)	Teammate 4 (Infra)	CRM, HR, Finance, Inventory, Support Module pages
Data Studio & Chart AI	Teammate 4 (Infra)	Teammate 2 (UI/UX)	`dataAnalystEngine.ts`, `CSVParser.tsx`, Chart Studio
RAG Knowledge Base	Teammate 4 (Infra)	Teammate 1 (AI)	`ragEngine.ts`, Knowledge Hub with citation search
Proactive Monitor & RBAC	Teammate 4 (Infra)	Teammate 3 (Workflow)	Proactive alert generator & RBAC security middleware